

Task 3: Docker Container for Website Monitor:

Dockerfile:

```
FROM eclipse-temurin:17-jdk

WORKDIR /app

COPY src/ ./src/

RUN mkdir -p bin && javac -d bin $(find src -name "*.java")

ENTRYPOINT ["java", "-cp", "bin", "Main"]
```

Docker Build and Output of Website-Monitor:

```
samuel@Samuel-PC:~/Dokumente/Uni/SoSe26/SWED/Ex4$ docker build -t website-monitor .
[+] Building 2.4s (9/9) FINISHED
=> [internal] load build definition from Dockerfile
=> => transferring dockerfile: 203B
=> [internal] load metadata for docker.io/library/eclipse-temurin:17-jdk
=> [internal] load .dockerignore
=> => transferring context: 2B
=> [1/4] FROM docker.io/library/eclipse-temurin:17-jdk@sha256:91b6210cce02091f6f0798a83ec51aa223828242c5a21a85793bb8c28dc891c4
=> => resolve docker.io/library/eclipse-temurin:17-jdk@sha256:91b6210cce02091f6f0798a83ec51aa223828242c5a21a85793bb8c28dc891c4
=> [internal] load build context
=> => transferring context: 1.44kB
=> CACHED [2/4] WORKDIR /app
=> CACHED [3/4] COPY src/ ./src/
=> CACHED [4/4] RUN mkdir -p bin && javac -d bin $(find src -name "*.java")
=> exporting to image
=> => exporting layers
=> => exporting manifest sha256:59384ed7df80567b101e5308027379fd8db7dfa5abddb555fcbcbaca94cc0a92
=> => exporting config sha256:4ef33cd6f9f63086e7b72381e74d864474e28fcfbab496823a47cfb8a678524e
=> => exporting attestation manifest sha256:58311c8e5896e0b6a2bd66b288cae20d873780c35dd89bf04679a4ef27303547
=> => exporting manifest list sha256:91fef0abd79b9c94b729d8b4a9da39517d3c89aecfd0bb55d8c2f93bbf3ae37f
=> => naming to docker.io/library/website-monitor:latest
=> => unpacking to docker.io/library/website-monitor:latest
samuel@Samuel-PC:~/Dokumente/Uni/SoSe26/SWED/Ex4$ docker run website-monitor "localhost"
WebsiteMonitor created with URL: localhost
samuel@Samuel-PC:~/Dokumente/Uni/SoSe26/SWED/Ex4$
```

Task 6: Exam Question:

Exam Question:

"a) Explain the core purpose of the Factory Method Pattern. Create a UML class diagram illustrating its structure and analyze how implementing this pattern affects the overall project architecture.

b) Compare the Factory Method Pattern with the Abstract Factory Pattern. Clearly state the structural and conceptual differences between them."

Mail to Prof. Lokaiczky:

